PRESIDENT OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
FORM FOUR NWZ MOCK EXAMINATIONS- 2025
BIOLOGY 2B MARKING SCHEME

1. (i) Procedures for preparation of solution R1 from specimen R

- Potato was sliced into small pieces.
- Sliced pieces of Potato were put into mortar and then grinded or crushed using pestle
- Little amount of water was added and then filtered to get potato solution.
- The solution was filtered in a beaker and labeled.

(ii) FOOD TEST REPORT

FOOD TESTED	PROCEDURES	OBSERVATION	INFERENCE
Starch **(0.5mark)**	To 2cm3 of solution R1 in a test tube, two drops of iodine solution were added then shaken **(01 mark)**	The mixture retained the colour of iodine solution **(01mark)**	Starch was absent **(0.5mark)**
Reducing sugar **(0.5mark)**	To 2cm3 of solution R1 in a test tube, 2cm3 of Benedict's solution was added then boiled **(01mark)**	The series of colour changed from blue, green, yellow, orange finally to brick red observed **(01mark)**	Reducing sugar was present **(0.5mark)**
Non reducing sugar **(0.5mark)**	To 2cm3 of solution R1 in a test tube, 1cm3 of diluted Hydrochloric acid was added then shaken and boiled for two minutes then cooled, followed by addition of 2cm3 of Sodium hydroxide and 2cm3 (equal amount) of Benedict's solution then boiled **(01mark)**	The series of colour change observed from blue, green, yellow, orange finally to brick red **(01mark)**	Non reducing sugar was present (**0.5mark)**
Protein **(0.5mark)**	To 2cm3 of solution R1 in a test tube, 2cm3 of sodium hydroxide solution was added, followed by two drops of 1% copper II sulphate solution, then shaken **(01mark)**	The blue colour of copper II sulphate solution was retained **(01mark)**	Protein was absent **(0.5mark)**

Lipids **(0.5mark)**	To 2cm3 of solution R1 in a test tube, 3 drops of Sudan III solution were added then shaken vigorously and left to settle for five minutes **(01mark)**	The solution form a red homogenous mixture with Sudan III solution **(01mark)**	Lipid was absent **(0.5mark)**

Total= 15marks

(iii) - Reducing sugar
- Non reducing sugar **(0.5@2 = 1mark)**

(iv) Glucose **(0.5mark)**

(v) Adaptation of ileum for absorption of digested food

- Presence of villus increases the surface area for food absorption
- The villi are well supplied with blood capillaries for transporting digested food
- The ileum is coiled to slow down movement of food
- The ileum has thin membranes for easy diffusion of food

(Any 4@1 = 4marks)

(vi) Part in alimentary canal is Ileum **(0.5marks)**

Enzymes used (0.5@2 = **(1mark)**

- **Maltase**
- **Lactase**
- **Sucrase**

(vii) They provide energy to the body (Sources of energy to the body) **TOTAL 25 MARKS**
Reducing sugars are the building blocks of complex sugars

(Any 2@0.5 = 1marks)

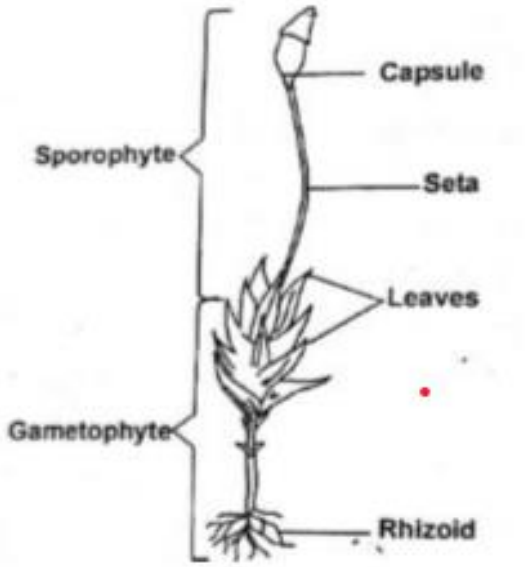
2. (a) G- Centipede
H- Tick
I- Bread mould **(0.5@ 6 = 3marks)**
J- Moss plant
K- Ascaris
L- Cactus plant

(b) Classification to class level

ORGANISM	KINGDOM	PHYLUM/DIVISION	CLASS
G	Animalia	Arthropoda	Chilopoda
H	Animalia	Arthropoda	Arachnida
I	Fungi	Zygomycota	Zygomycetes
J	Moss plant	Brayophyta	Musci/ mosses
K	Animalia	Nematoda	chromadorea

(0.5@ 15= 7.5marks)

c) Diagram of moss plant



Caption = 0.5marks

Labeling 4 parts@1 = 4 marks

Diagram = 1marks

(Total= 05.5marks)

(d) Habitat of specimen H and J

H- Land on the skin of animals like cows and dogs **(1mark)**

J- Land or terrestrial habitat (**moist area**) **(1mark)**

(e) **Adaptation of cactus to its habitat**

- They have spiny leaves for reducing transpiration or water loss
- They have thick and fleshy stem for storage of water
- They have sunken stomata which minimize losing water by transpiration
- They have many root hairs for collecting and absorption of droplets of water
- They have waxy waterproofing coat for storage of water in the cactus
- Cactus have deeper tap root for anchoring the cactus and absorption of water.

(Any 4@ 1= 4marks)

(f) **Advantages of cactus in daily life**

- Cactus are used to prevent soil erosion
- Used to make drugs
- They provide oxygen in the atmosphere which is used by animals.
- Provide shelter and shadow to small animals **(Any 3@1=3marks)**
- They are used in scientific investigation and study

TOTAL 25 MARKS